

Strings :- sequence of characters

OR

whatever we put inside double quotes.

Ex-

"Kiaan"

"Azaan"

"24"

" / a # @ \$ - Kiaan "

→ strings are immutable → which cannot change after creation

## Data Types

Imagine a bank account.

A bank account contains different kinds of information:

- Account Holder Name → "Shanmuka"
- Account Balance → 50000.75
- Account Active → True
- Transaction IDs → [101, 102, 103] → collection

All these values are different.

Python needs to know:

- Which value is text?
- Which value is a number?
- Which value is a yes/no answer?
- Which value is a collection?

This is where **Data Types** become important.

variables

↑

What kind of data it stores →

Variables:-

name = "Sophie" ←

age = 16 ←

wt = 45.75 ↘

## Data Types

Single Value

Collection Data Types

These store one value at a time.

Examples:

- int ✓ → age → 12
- float ✓ → wt → 42.35
- str ✓ → name → "Tejas"
- bool ✓ → active → True

not can change

These store multiple values together.

Examples:

- list
- tuple

you can access index

- int ✓ → age → 12
- float ✓ → wt → 42.35
- str ✓ → name → "Tejas"
- bool ✓ → isactive → false

that can change

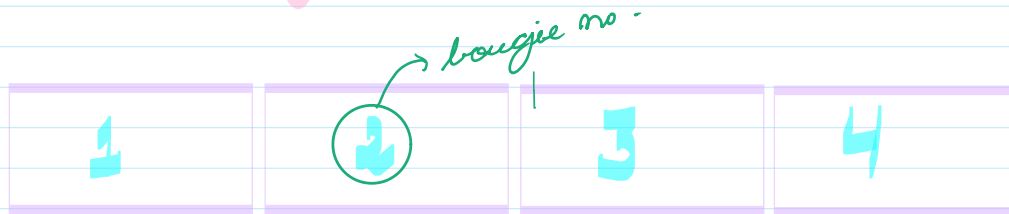
Examples:

- list
- tuple
- set
- dict

you can access by Index

|            | Data Type | Example           | Mutable/Immutable | Ordered/Unordered | Real-Life Example |
|------------|-----------|-------------------|-------------------|-------------------|-------------------|
| single     | int       | 100               | Immutable         | N/A               | Roll Number       |
|            | float     | 99.5              | Immutable         | N/A               | Product Price     |
|            | str       | "Python"          | Immutable         | Ordered           | Student Name      |
|            | bool      | True              | Immutable         | N/A               | Login Status      |
| collection | list      | [1,2,3]           | Mutable           | Ordered           | Shopping Cart     |
|            | tuple     | (1,2,3)           | Immutable         | Ordered           | Coordinates       |
|            | set       | {1,2,3}           | Mutable           | Unordered         | Unique IDs        |
|            | dict      | {"name": "Rahul"} | Mutable           | Insertion-Ordered | Student Record    |

# Indexing:-



TRAIN

K I A A N

indexing → 0 1 2 3 4

Why indexing starts with 0?

Index= How much in terms of distance a particular character is differ from first character.